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Development of Innovative Polymers as Cement Retarders: Lab Study and Application

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Abstract

Cement must be designed in a way to ensure acceptable properties such as mixability, stability, rheology, fluid loss, and adequate thickening time. Different chemicals are used when designing cement slurries. Retarders are cement additives whose function is to retard, or delay the setting of cement slurries. For a well whose temperature is about 125 °F or less, we do not need to add any retarders. However, as temperature increases, the hydration process of Tricalcium silicate (C3S) increases and, hence, the thickening time decreases. In addition to retarder additives which are added to cements, retarded cements are available. These contain lignin, gums, starches, etc. which provide the retarding action. However, these may not be compatible with cement additives and the setting times may be difficult to control consistently. To explain the mechanism of retardation in Portland cement, the following four theories were proposed: adsorption, precipitation, nucleation and complexation. According to the adsorption theory, the retarder is absorbed on the surface of the hydration products inhibiting contact with water. This theory suggests that retardation is due in part to the adsorption onto the surface of the C-S-H gel hydration product formed around the grains rendering it hydrophobic. The precipitation theory suggests that the retarder reacts with calcium and/or hydroxyl ions and forms an impermeable sheet covering cement grains. In the nucleation theory, it is suggested that the retarder slows down the growth rate of hydration products by adsorbing on their nuclei. Finally, the complexation theory states that calcium ions are chelated by the retarder preventing the formation of the nuclei. It is possible that all of the above effects are involved to some extent in the retardation process. The predominant factor depends on the type of retarder used and the cement phases upon which the retarder acts.

This study will detail mechanisms, interactions and lab and deployment performance of different polymers functions as retarders. The study will include thickening time tests, rheology tests, compressive strength development and compatibility tests with other conventional additives. Polymers with hydrocarboxylic acid derivatives and polymers with sodium lignosulfonate and/or hydrolyzed carbohydrates were fully investigated and characterized using proton NMR and FTIR techniques. These polymers were also used in effective formulations to cement different casings successfully.

Introduction

Oil well cementing is a vital process during drilling and completion. It involves injecting cement slurry between the casing and the formation, ensuring long-term zonal isolation [Magarini, 1999]. Cementing operations are typically classified into primary and secondary types. Primary cementing occurs right after drilling and aims to secure the casing, offering essential protection and isolating the formation from the wellbore.

The design of primary cement slurry demands careful attention, as the quality of primary cementing significantly impacts the well's longevity. When selecting additives for this process, consider various factors such as well profile, temperature, pressure, and the presence of acidic gases. After the cement slurry is formulated, it is pumped into the well and positioned between the casing and the geological formation. The setting time depends on the types of retarders and accelerators added to the slurry.

The main purpose of primary cementing is to support and bond the casing, ensure zonal isolation to prevent fluid migration between formations, protect the casing from corrosive environments, serve as a seal and barrier to contain formation pressure, and shield the casing from drilling impacts such as shock loads. Casing points and cementing stages are typically outlined in the well design, making the driller aware that drilling must be scheduled accordingly for casing and cementing. Nonetheless, there are situations where casing and cementing become necessary to resolve specific drilling issues, such as lost circulation. Successful cementing begins with designing effective cement slurries, often by adding additives to enhance or modify properties to meet desired objectives [Anon, 2014].

The additives included in cement are a crucial part of well design, construction, and integrity. Therefore, comprehensive lab testing must be conducted for any new cement formulation or additive before it is used in the cement slurry. Lab testing includes compression tests using UCA, setting time, and more. Several additives are incorporated into cement design for specific applications [Sauer, 1985].

Selecting additives for cement design depends on formation type, depth, temperature, and pressure. In addition to standard additives used for rheology, fluid loss control, and retarders, specialty additives like extenders, thinners, gas migration preventers, H₂S and oxygen scavengers, corrosion inhibitors, and expansion agents should be chosen based on well requirements. Properly designing the cement slurry for each job can also mean excluding unnecessary, costly chemicals to reduce overall cementing costs. Since each well has unique characteristics, cementing slurries must be tested beforehand. This process requires significant experience from cementing specialists to assist drilling engineers in crafting effective cement slurries.

It is the process of placing a cement slurry between a casing or liner string and the drilled formation. The main objectives of this operation are to include zonal isolation to prevent fluid migration in the annulus, provide support for the casing or liner string, and protect the casing from corrosive formation fluids. The cement slurries are routinely designed for temperatures ranging from below freezing in permafrost zones to 700°F in geothermal and thermal recovery wells, with pressures varying from near ambient in shallow wells to well over 30000 psi in deep wells. Besides temperature and pressure, other factors such as porous formations, corrosive fluid environments, and overpressured formations must also be considered in primary cement design. Only by incorporating adequate additives into the cement slurry can such a wide range of conditions be accommodated. Additives enable easy and successful placement, rapid development of compressive strength, and provide zonal isolation for the lifetime of the well. The abbreviations listed in Table 1 are used to simplify the expression of different cement chemical constituents.

Table 1—abbreviations to simplify the expression of different cement chemical constituents

$C=CaO$; $F=Fe_2O_3$; $N=Na_2O$; $P=P_2O_5$; $A=Al_2O_3$; $M=MgO$; $K=K_2O$; $F=FeO$; $S=SiO_2$; $H=H_2O$; $L=Li_2O$; $T=TiO_2$; $S=SO_3$; $C=CO_2$

Portland cement mainly consists of four components: tricalcium silicate (C3S), which contributes to early strength; dicalcium silicate (C2S), responsible for final strength; tricalcium aluminate (C3A); and tetracalcium alumina ferrite (C4AF), both of which aid early strength. Gypsum is added to regulate the cement's setting time [Al-Yami, 2007]. The silicate phases account for about 75-80% of the total cement, with C3S being the predominant (60-65%). C2S usually makes up less than 20%, except in retarded cements. Upon hydration, C3S and C2S generate calcium silicate hydrate (C3S2H) and calcium hydroxide (Ca(OH)₂), also called Portlandite. The C3S2H₃, known as CSH gel, varies in C:S and H:S ratios depending on temperature, calcium levels, and curing duration. This gel constitutes approximately 70% of fully hydrated Portland cement at room temperature and is the main binder in hardened cement. Conversely, calcium hydroxide is highly crystalline, representing about 15-20%, and is responsible for the high pH of cement.

When water contacts gypsum, it partially dissolves, releasing calcium and sulfate ions that react with aluminate and hydroxyl ions from C3A to produce calcium trisulfoaluminate hydrate, commonly known as ettringite (C3AC3S32H). This compound precipitates onto C3A surfaces, preventing rapid hydration or flash-set. As gypsum gradually depletes, ettringite continues to form until all gypsum is consumed. The sulfate level then drops, causing ettringite to become unstable and convert into calcium monosulfoaluminate hydrate. The remaining unreacted C3A forms calcium aluminate hydrate. Cement formulations often include specific additives to modify or inhibit hydration reactions [Nelson, 1990]. For the same slurry composition, outcomes can vary widely based on factors such as particle size distribution, silicate and aluminate phase distribution, reactivity of hydrating phases, gypsum/hemihydrate ratio, total sulfate content, free alkali levels, temperature, pressure, additive concentrations, mixing energy, mixing sequence, and water-to-cement ratio. Due to the complexity of cement hydration and the influence of many parameters, thorough laboratory testing is essential before field application to prevent unexpected issues. Over a hundred additives—liquid and solid—are available in the market, categorized into eight main application groups.

Retarders are chemicals that prolong the setting time of cement. Although their exact mechanism remains unclear, several theories have been proposed in the literature. The effectiveness of a retarder depends mainly on its chemical nature and the cement phase. For wells at about 125 °F or lower, no retarders are necessary. However, increasing temperature speeds up the hydration of C3S, reducing thickening time. Retarders can be added during cement mixing or used in retarded cements available commercially. Four main theories explain retarder action: adsorption, precipitation, nucleation, and complexation [Nelson, 1990]. The adsorption theory suggests retardation occurs when the retarder adsorbs onto hydration product surfaces, hindering water contact. Precipitation involves the retarder reacting with calcium or hydroxide ions, forming an insoluble, impermeable layer around cement particles. In the nucleation theory, the retarder adsorbs onto hydration nuclei, acting as a poison and preventing growth. The complexation theory describes calcium ions chelated by the retarder, inhibiting nucleus formation. Multiple mechanisms may operate simultaneously. The chemical nature of retarders is well understood and summarized in the following section.

Sodium and calcium salts of lignosulfonates, as shown in Fig.1, are the most common retarders used in oil well cementing. These are unrefined wood polymers containing various amounts of saccharide compounds. Their molecular weight typically ranges from 20000 to 30000. The presence of low molecular weight carbohydrates contributes to the retardation effect of lignosulfonates, as noted in [Milestone, 1979].

For effective retardation, the recommended dosage ranges from 0.1% to 1.5% BWOC. Lignosulfonates influence the hydration kinetics of C3S through adsorption and nucleation mechanisms. Additionally, their retardation effect may result from changes in the morphology of the C-S-H gel structure, where adsorption of lignosulfonate forms an impermeable layer that inhibits further hydration, as described by Jennings et al., 1986.

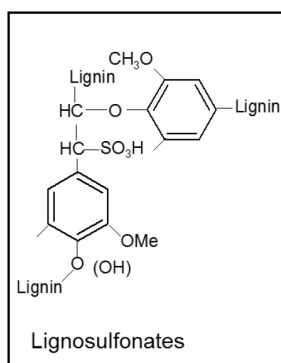


Figure 1—Lignosulfonates

The retarding effect of hydroxycarboxylic acids and their salts is generally due to the presence of alpha or beta hydroxycarboxylic acid groups, as illustrated in Fig. 2. These groups can chelate metal ions such as calcium, forming a stable ring structure. This chelation partially adsorbs on the hydrated cement surface and inhibits the nucleation sites of hydration products. Gluconate and glucoheptonate salts are the most commonly used retarders. Additionally, citric acid is employed as a retarder and also acts effectively as a cement dispersant.

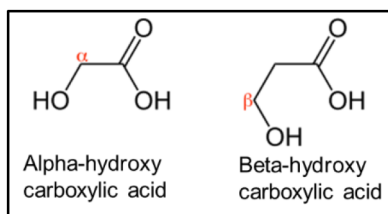


Figure 2—Hydroxycarboxylic acids

For saccharide compounds, the most effective retarders are those with five-membered ring structures, such as sucrose and raffinose, as shown in Fig. 3 [Thomas and Birchall, 1983]. However, because of their high sensitivity to small concentration changes, they are not commonly used in well cementing. It has been reported that their retardation effect depends on the susceptibility of these compounds to alkali hydrolysis, which inhibits hydration by poisoning nucleation sites [6].

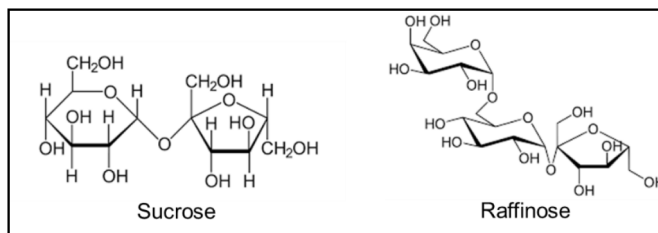


Figure 3—Saccharide compounds

Cellulose derivatives are polysaccharides obtained from wood and remain stable in basic cement slurry conditions. Their retardation mechanism involves these cellulose compounds adsorbing onto the hydrated cement surface, thereby blocking further hydration. The active sites responsible for this adsorption are the ethylene oxide links and carboxyl groups within the cellulose. The most commonly used cellulosic retarder is carboxymethyl hydroxyethyl cellulose, as shown in Fig. 4 [Shell and Wynne, 1958]. It is frequently employed as an additive to control fluid loss in cement.

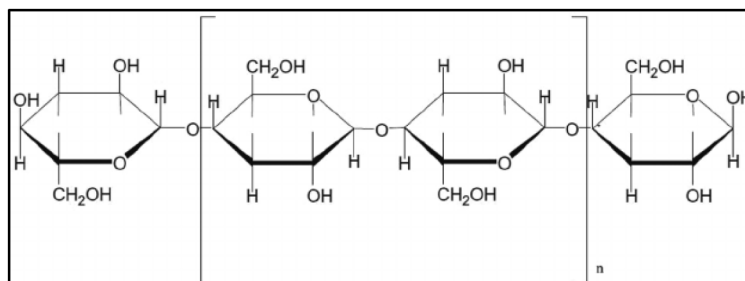


Figure 4—Carboxymethylhydroxyethylcellulose

The insensitivity of organophosphates with the general formula shown in Fig. 5 to minor variations in cement composition is a beneficial property that makes these compounds effective as retarders. Alkylene phosphonic acids and their salts have been identified as retarders in oil well cementing, demonstrating excellent hydrolytic stability and effectiveness at bottom hole temperatures up to 400°F [Nelson, 1987]. Additionally, phosphomethylated compounds containing quaternary ammonium salts are highly efficient retarders [Crump and Wilson, 1984]. Although the exact mechanism of retardation is not well understood, it is believed that, similar to other retarder classes, their action involves poisoning active nuclei through adsorption, which retards the setting time.

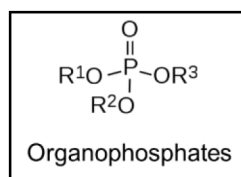


Figure 5—Organophosphates as retarders

Other inorganic acids like boric acid, phosphoric acid, hydrofluoric acid, and chromic acid, along with their salts, can delay the hydration of Portland cement. Oxides of zinc and lead are also used as retarders for thixotropic cement slurries. Sodium chloride at concentrations above 20% BWOW has been employed as a cement retarder. Besides retarders, some chemicals like sodium tetraborate decahydrate serve as retarder aids. This compound can increase the temperature limit of lignosulfonate to 600°F. However, adding sodium tetraborate decahydrate can reduce the effectiveness of cellulosic and polyamine fluid loss additives.

Experimental Studies

Slurry Preparation Procedure

The slurry formulation was prepared in the lab using a standard API blender. It was mixed for 15 seconds at 4,000 rpm, then for 35 seconds at 12,000 rpm, with the maximum speed reaching 12,000 rpm during preparation.

Slurry Rheology

The slurry was conditioned in an atmospheric consistometer prior to taking rheological measurements. An Fann viscometer (Model-35) was used to determine the slurry's apparent viscosity.

Thickening Time Test

The prepared slurry was poured into the API standard HP/HT consistometer slurry cup for the thickening time test, which is essential to assess the pumpability of the cement slurry.

Fluid Loss Test

Dynamic fluid loss, occurring during slurry placement, causes increased slurry density, potentially leading to loss circulation. It can also negatively impact properties like rheology and thickening time. Conversely, static fluid loss happens after placement and, if not managed properly, can reduce slurry volume and drop pressure, risking formation fluids entering the slurry.

Results and Discussion

Three retarders were tested to identify suitable operational ranges for their effects on cement slurries. Retarder 1 (Fig.6) appears as a dark brown liquid primarily composed of Sodium lignosulfonate. Its alkali content is less than 5.0g of Na₂O equivalent per liter of admixture. The retardation mechanism of lignosulfonate is generally believed to involve a combination of adsorption and nucleation processes. Sodium lignosulfonate contains the functional groups responsible for delaying the cement hydration reaction.

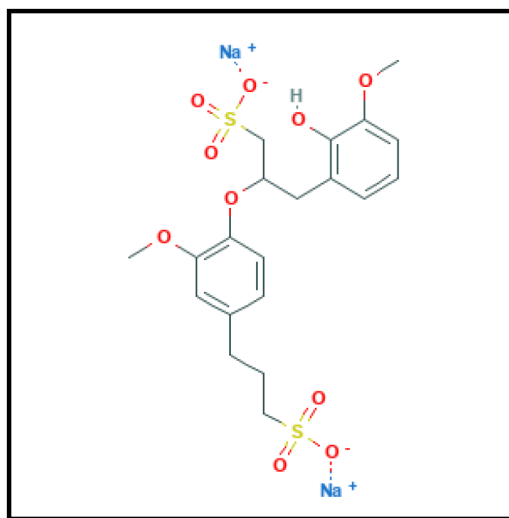


Figure 6—Sodium lignosulfonate chemical structure C₂₀H₂₄Na₂O₁₀S₂

Retarder 2 is a dark brown liquid primarily composed of Sodium lignosulfonate and hydrolyzed carbohydrates (see Fig. 7). Carbohydrates, also known as sugars or saccharides, are compounds that have been broken down into their simplest form through hydrolysis, such as sucrose being hydrolyzed into glucose and fructose. The alkali content of Retarder 2 is less than 4.0 g of Na₂O equivalent per liter of mixture. Various concentrations were tested to determine the effective range for both retarder 1 and retarder 2. The optimal concentrations for each were chosen for low-temperature cementing applications, as shown in Table 2 and Figures 8-9.

Table 2—Acceptance Criteria for R1 retarder and R2 retarder

R-1(118 PCF) Class G Cement +0.002gps Defoamer +0.06gps R1 or R2					
Temp. (degF)	Conc. (gps)	TTRB (min)	BHP (psi)	TT (hr:mn)	Acceptance Criteria
150	0.06	45	5600	> 3: 30	If Thickening time is less than 3.5 hours, increase concentration to 0.08 gps and run again. If TT still not achieved, reject this batch
				3:56	Passed

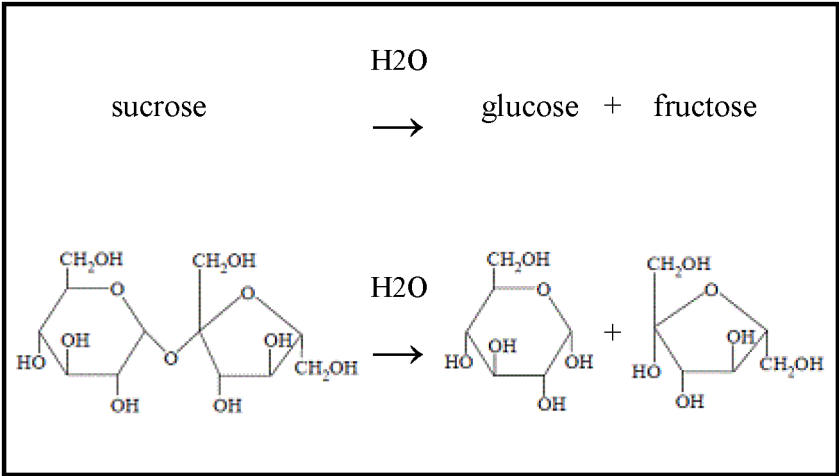


Figure 7—liquid mainly consist of combination of Sodium lignosulfonate and hydrolyzed carbohydrates

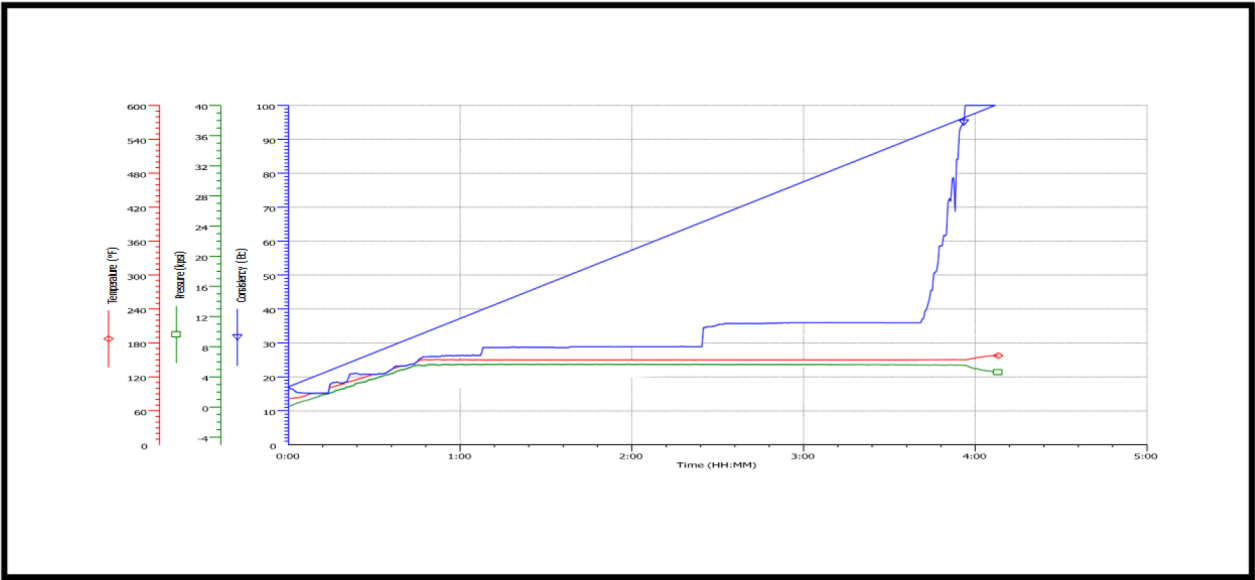


Figure 8—R1 thickening time chart for low temperature range

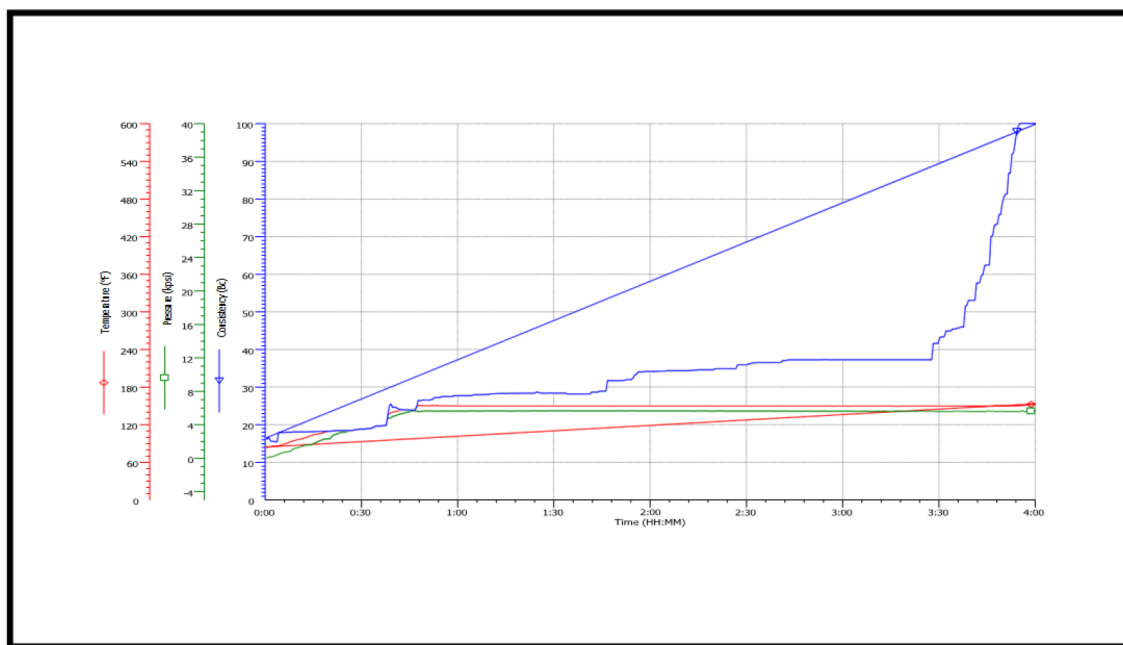


Figure 9—R2 thickening time chart for low temperature range

Retarder 3 (Fig.10) is a brown, chloride-free liquid based on a Hydrocarboxylic acid derivative, primarily composed of Sodium gluconate, which is the sodium salt of gluconic acid. Its chemical formula is $\text{NaC}_6\text{H}_{11}\text{O}_7$. Hydrocarboxylic acids contain hydroxyl and carboxyl groups in their molecular structures. Sodium gluconate is a commonly used material in this category due to its strong retarding properties and potential to cause over-retardation at BHCT below 200°F. The retardation effect of Hydrocarboxylic acids and their salts is mainly due to alpha or beta Hydrocarboxylic groups ($\text{HO-C-CO}_2\text{H}$ and $\text{HO-C-C-CO}_2\text{H}$, respectively) that can chelate metal cations like calcium effectively. The retardation mechanism for Hydrocarboxylic acids is explained by nucleation theories. The optimal concentrations for both retarders were chosen for low-temperature cementing applications, as shown in Table 3 and Fig.11.

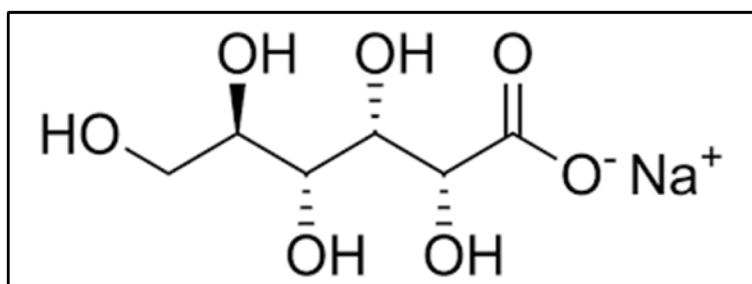


Figure 10—Chemical structure for sodium gluconate

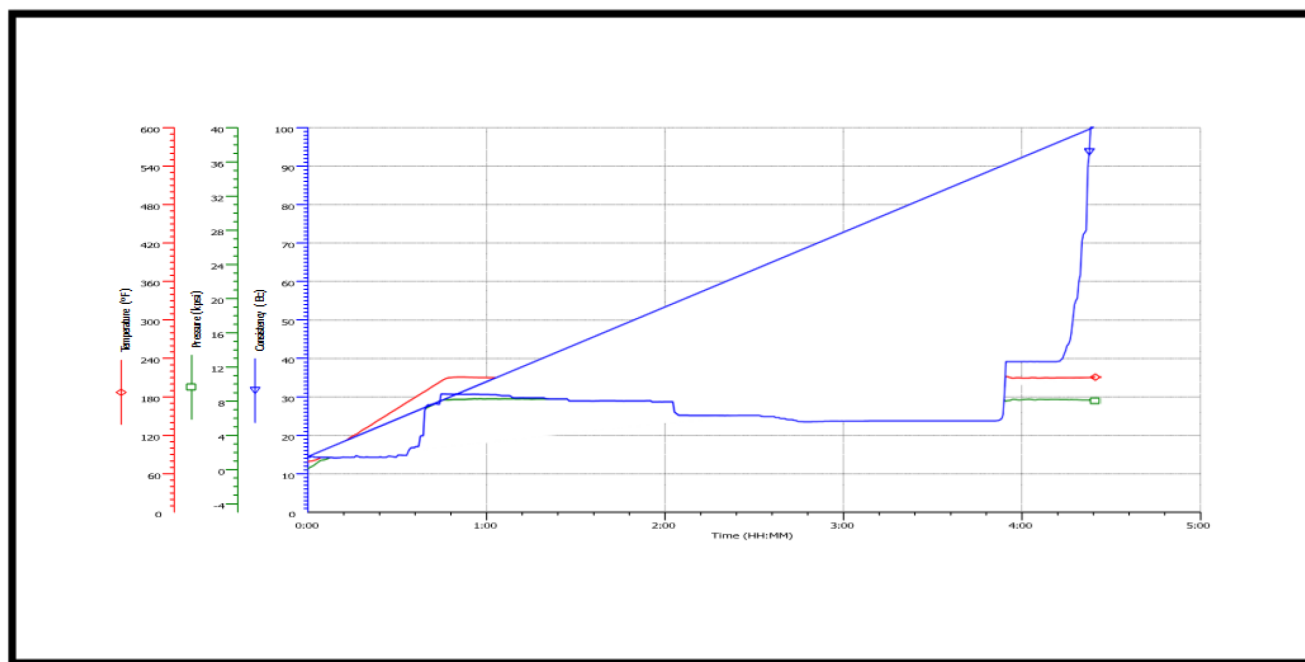


Figure 11—R3 thickening time chart for low temperature range

Table 3—Acceptance Criteria for R3 retarder

R-1(118 PCF) Class G Cement+ 35% BWOC Silica Flour +0.002gps Defoamer+0.05gps R3					
Temp. (degF)	Conc. (gps)	TTRB (min)	BHP (psi)	TT (hr:mn)	Acceptance Criteria
210	0.05	45	8000	> 3: 30	If Thickening time is less than 3.5 hours, increase concentration to 0.06 gps and run again. If TT still not achieved, reject this batch
				4:23	Passed

Deployment of Retarder in cement slurry

Retarder 3 was used in the second-stage lead and tail cement slurry for cementing 185/8" casing, as shown in Fig.12.

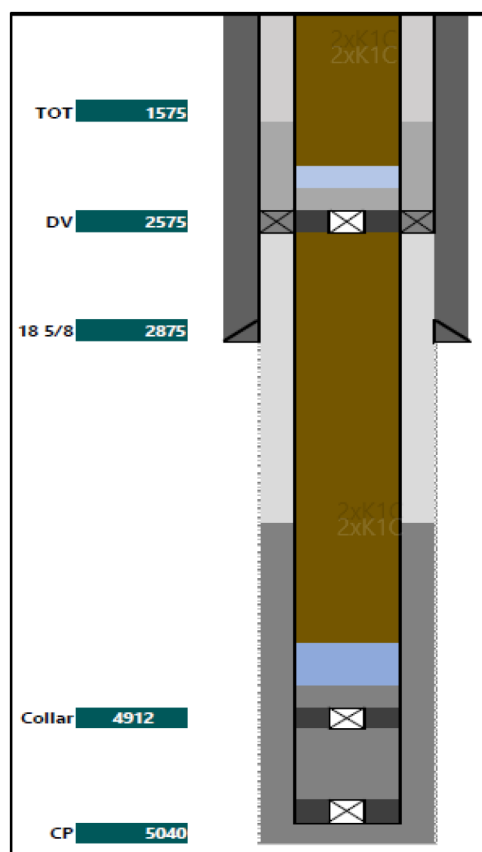


Figure 12—Schematic of the well in which the retarder was deployed

The retarder was added in the second stage of the 101 pcf cement slurry. Table 4 shows the formulation of the 101 pcf cement slurry.

Table 4—2nd stage lead 101pcf cement slurry

Additives	Concentration	Unit
Dry Blend		
Class G cement	100	%BWOC
Mix Fluid		
Defoamer	0.005	gps
Extender	1.8	%BWOC
Gelling agent	0.25	%BWOC
Novel Retarder	0.03	gps

The rheology of the 101 pcf cement slurry was measured at both room temperature and after conditioning the slurry at 105°F. Table 5 presents the rheological data for the 101 pcf second stage lead slurry.

Table 5—Rheology Data of 101pcf 2nd stage lead slurry

Speed rpm	Readings @ RT			Readings @ 105°F		
	up	down	Avg	up	down	Avg
300	101	101	101	79	79	79
200	81	80	80.5	52	51	51.5
100	44	42	43	31	30	30.5
60	34	33	33.5	26	25	25.5
30	31	20	20.5	15	15	15
6	12	12	12	11	11	11
3	11	11	11	10	10	10
	10sec gel	10min gel	1min stirring	10sec gel	10min gel	1min stirring
	10	30		10	15	
	PV, cp		93.3	PV, cp		67.6
	YP, (lb/100ft ²)		12.7	YP, (lb/100ft ²)		9.8

Thickening time experiment was performed on the 101pcf 2nd stage lead slurry. Fig. 13 shows the thickening time chart for the 101pcf cement slurry with the novel retarder.

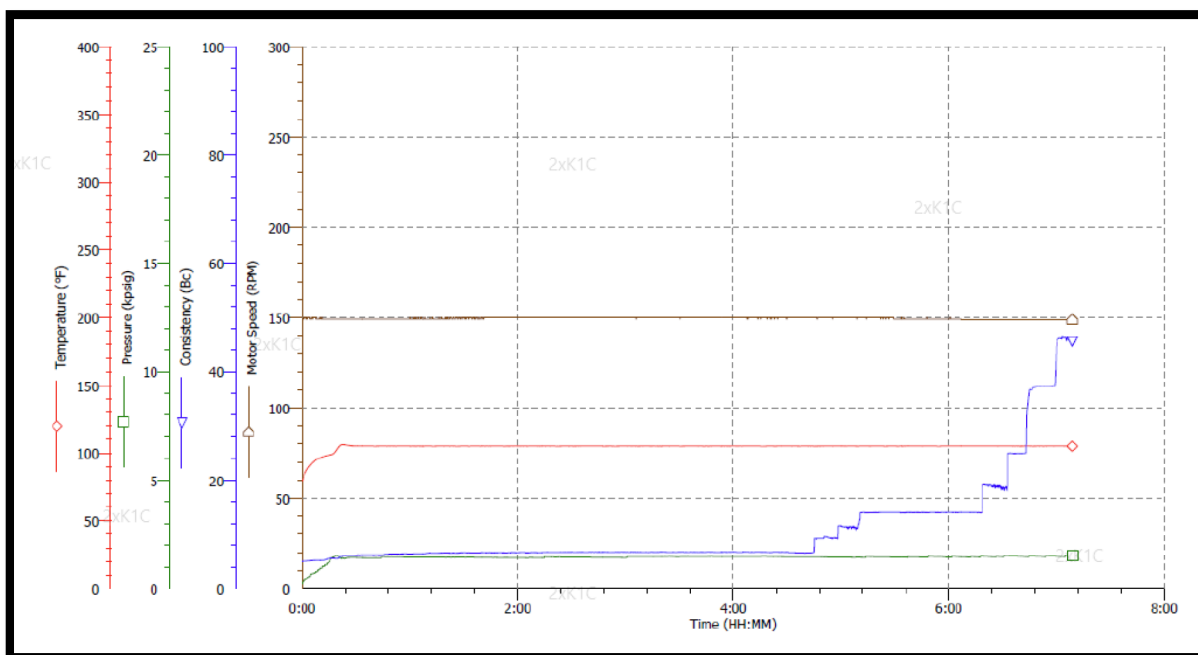


Figure 13—Thickening time chart for the 101pcf cement slurry with the novel retarder

The retarder was applied in the second stage of the tail 118 pcf cement slurry. Table 6 provides the formulation for the 118 pcf cement slurry.

Table 6—2nd stage tail 118pcf cement slurry

Additives	Concentration	Unit
Dry Blend		
Class G cement	100	%BWOC
Expansion agent	1	%BWOC
Mix Fluid		
Defoamer	0.005	gps
LT Anti-settling agent	0.10	%BWOC
Fluid loss agent	0.40	%BWOC
Dispersant	0.05	%BWOC
Novel Retarder	0.015	gps

The rheology of the 118 pcf cement slurry was measured at both room temperature and after conditioning the slurry at 105°F. Table 7 presents the rheology data for the 101 pcf second stage lead slurry.

Table 7—Rheology Data of 118pcf 2nd stage tail slurry

Speed rpm	Readings @ RT			Readings @ 105°F		
	up	down	Avg	up	down	Avg
300	108	108	108	97	97	97
200	82	81	81.5	69	71	70
100	51	50	50.5	40	42	41
60	39	38	38.5	30	30	30
30	32	31	31.5	20	19	19.5
6	14	13	13.5	11	11	11
3	11	10	10.5	10	10	10
	10sec gel	10min gel	1min stirring	10sec gel	10min gel	1min stirring
	11	49	24	11	20	
	PV, cp		95.1	PV, cp		87.5
	YP, (lb/100ft ²)		17.0	YP, (lb/100ft ²)		11.3

The thickening time experiment was conducted on the 118 pcf 2nd stage tail slurry. Fig. 14 presents the thickening time chart for the cement slurry with the new retarder.

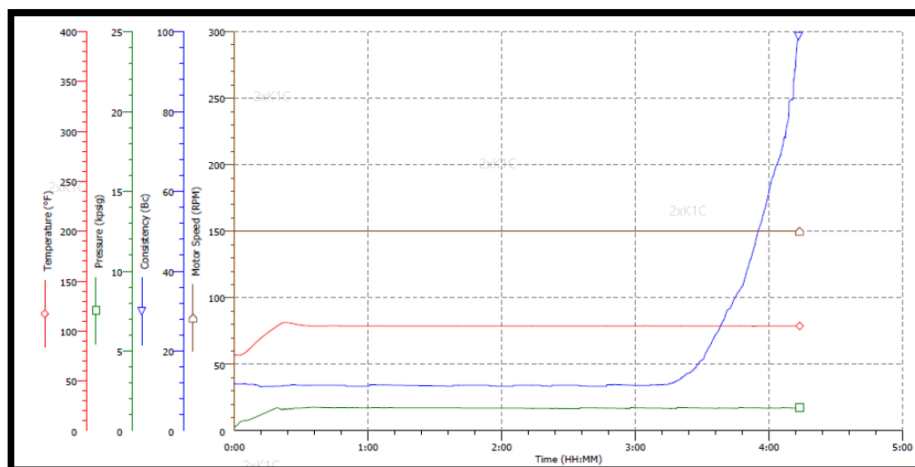


Figure 14—Thickening time chart for the 101pcf cement slurry with the novel retarder

The slurry experienced a thickening time of 4 hours and 13 minutes (@100Bc) at a final temperature of 105°F and a final pressure of 1480 psi. Fig. 15 illustrates the process in terms of density, pressure, and flow rate. The casing was cemented successfully, with no issues during pumping, and the retarder fulfilled its goal of safe placement.

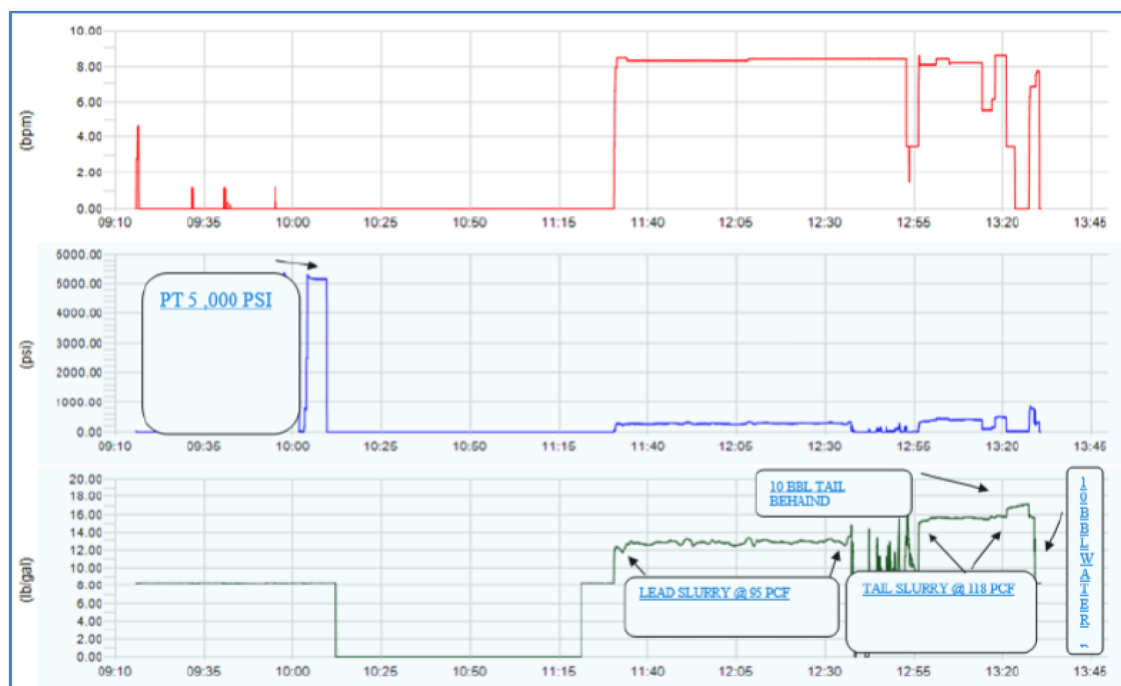


Figure 15—Thickening time chart for the 101pcf cement slurry with the novel retarder

Conclusions

- Three retarders were studied to define their effective working range for retarding cement slurries.
- Concentrations of sodium lignosulfonate and hydrolyzed carbohydrates retarders were optimized for low-temperature cementing.
- Sodium gluconate retarder concentration was optimized for medium-temperature cementing.
- The field application using sodium gluconate was successfully carried out.

Abbreviations

API	: American Petroleum Institute
Bc	: Bearden consistency
BHCT	: Bottom Hole Circulating Temperature, °F
BHST	: Bottom Hole Static Temperature, °F
BP	: British Petroleum
BV	: Bulk Volume, inch ³
BWOC	: By weight of Cement
DP	: Differential Pressure, psig
ECD	: Equivalent Circulating Density, lb. /gal
HT/HP	: High Temperature/High Pressure
ID	: Inside Diameter, inch
OD	: Outside Diameter, inch

PV : Plastic Viscosity, cp
 PCF : Pounds per cubic foot
 RP : Recommended Practices
 TD : Total Depth, ft.
 TDS : Total Dissolved solids, mg/l
 WOC : Waiting on Cement
 YP : Yield Point, lb. /100 ft.

SI Metric Conversion Factors

in. $\times 2.54^*$	E-02 =m
(°F-32) / 1.8 $\times 10^0$	= °C
Ft $\times 3.048 \times 10^{-1}$	= m
Gal $\times 3.785412 \times 10^{-3}$	= m ³
lbm $\times 4.535924 \times 10^{-1}$	= kg
Psi $\times 6.894757 \times 10^{-3}$	= Mpa
lbm/gal $\times 1.19826 \times 10^{-1}$	= S.G
bbl. $\times 1.58987 \times 10^{-1}$	= m ³

*Conversion factor is exact.

References

- Al-Yami, A.S., Al-Arfaj, M.K., Nasr-El-Din, H.A., Jennings, S., Khafaji, A., Al-Ariani, M. and Al-Humaidi, A. Development of New Retarder Systems To Mitigate Differential Cement Setting in Long Deep Liners. SPE/IADC 107538, 2007, Middle East Drilling Technology Conference & Exhibition, Cairo, Egypt.
- Anon, American Petroleum Institute (API) Recommended Practice 10B for Testing Well Cements. American Petroleum Institute Washington, D.C., USA, 2014, 134.
- Crump, D.K. and Wilson, D.A. Set-Retarding Additives for Cement from Aminomethylene phosphonic acid Derivatives. U.S. Patent No. 4468252, 1984.
- Jennings, H.M., Taleb, H., Frohnsdorff, G. and Clifton, J.R. Interpretation of the Effects of Retarding Admixtures on Pastes of C3S, C3A plus Gypsum and Portland Cement. Proc. Eighth Intl. Cong. Chem. Cement, Rio de Janeiro. 1986, 239–243.
- Magarini, P., Lanzetta, C., and Galletta, A. Drilling Design Manual Eni-Agip Division. 1999, 97–107.
- Milestone, N.B. Hydration of Tricalcium Silicate in the Presence of Lignosulphonates. *Glucose and Sodium Gluconate*. Amer. Ceramic Soc. 1979, **62**, 321–324.
- Nelson, E.B. Well Cementing, Schlumberger Educational Services, Houston, 1990.
- Nelson, E.B. Well Treating Process and Composition. Canadian Patent No. 1216742, 1987.
- Sauer, C.W. and Landrum, W.R. Cementing-A Systematic Approach, *Journal of petroleum technology*, 1985, 2184–2196.
- Shell, F.J. and Wynne, R.A. Application of Low-Water-Loss Cement Slurries. paper API 875-12-I, 1958.
- Thomas, N.L. and Birchall, J.D. The Retarding Action of Sugar on Cement Hydration. *Cement and concrete research*. 1983, **13**, 830–842.